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Improving collaborative filtering-based recommender systems results using Pareto dominance

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ABSTRACT

Recommender systems are a type of solution to the information overload problem suffered by users of websites that allow the rating of certain items. The collaborative filtering recommender system is considered to be the most successful approach, as it makes its recommendations based on ratings provided by users who are similar to the active user. Nevertheless, the traditional collaborative filtering method can select insufficiently representative users as neighbours of the active user. This means that recommendations made *a posteriori* are not sufficiently precise. The method proposed in this paper uses Pareto dominance to perform a pre-filtering process eliminating less representative users from the *k*-neighbour selection process while retaining the most promising ones. The results from experiments performed on the Movielens and Netflix websites show significant improvements in all tested quality measures when the proposed method is applied.

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1. Introduction

Recommender Systems (RS) are responsible for providing users with a series of personalised suggestions (recommendations) for certain types of items. RS extract the user's relevant characteristics and determine the subset of items that may be of interest to them.

The way RS work can differ to a great extent depending on the types of items to be recommended and the available information about the user's preferences. Therefore, RS must be able to use different mechanisms to obtain the most promising items for each user. RS are commonly called "filters" because they act as such. Currently, the following types of RS are found in practice [1,12]:

- Content-based filtering [3,26]: Recommendations are based on preferences users have expressed in the past about the items (e.g., books purchased).
- Demographic filtering [22]: Recommendations are based on positive ratings from other users who share the same age, geographical location, gender, profession, etc. as the active user.
- Collaborative filtering (CF) [1,8–10,17,20,23]: Relevant information is stored in a database that contains the ratings of a large number of users on a large number of items (films, books, jokes, study material, holiday destinations, etc.). CF aims to determine which users are similar to each other and then to recommend to the active user those items preferred by users similar to him or her.
- Hybrid filtering methods [2,4,11,12,14]: In some cases, content-based filtering and demographic filtering complement CF. By combining these methods with CF, recommendations can be made based on a more extensive set of information.

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Generally speaking, CF obtains better results than the other filtering systems described [12,16]; thus, it is the most widely used filtering method. However, CF suffers from two main interrelated problems: (1) It requires a large enough database to guarantee the correct process to search for similar users and to make the recommendations [17]. (2) The rating matrix is typically excessively sparse [24], which makes it difficult to calculate the similarity between pairs of users and reduces the accuracy of the computed recommendations.

In this paper, we present a method that improves CF prediction and recommendation quality measures. The rest of the paper is structured in the following way:

- In Section 2, we describe the motivation for and the foundations of the proposed method.
- In Section 3, we formalise the process by which recommendations are obtained.
- In Section 4, we define experiments designed to validate the correct functioning of the proposed method.
- In Section 5, we compare results obtained by applying the new method to results obtained by applying the traditional method.
- In Section 6, we explain our conclusions and propose future work.

2. Designing the new method

2.1. Introduction

The main objective of CF is to provide an user, whom we will call the active user, with a series of suggestions about items that may be of interest to him or her. Its operation is based on a very simple idea [1]: if two users have similar preferences, it is highly probable that one of them will like items they are not aware of if the other user likes them. Therefore, the purpose of CF is to search for those users who are most similar to the active user and then analyse their ratings to determine which items may interest him or her. This process can be summarised in the following three steps [1,9,16]:

1. Find the k users most similar to the active user (the k-neighbours of the active user). This phase has the most significant impact on the quality of the recommendations. The method proposed in this paper provides a novel approach for obtaining a suitable set of neighbours to the active user.
2. Predict the rating that the active user would give to items they have not yet rated, by observing the ratings of their k-neighbours. When trying to predict an item's value, there will normally be a significant number of neighbours who have not rated the item; therefore, mechanisms must be defined that enable the k-neighbours' ratings to be combined satisfactorily.
3. Find the most suitable N items to be recommended (due to their high rating, novelty, etc.).

2.2. Motivation

One of the main problems faced by CF is the high degree of sparsity [24] in the rating databases, which arises from the small percentage of available items for which a given user generally provides ratings. Thus, when we want to calculate the similarity between each pair of users, we must do so by only considering the items that both users have rated in common. Traditional similarity metrics [1,12], including Pearson Correlation, Cosine, and Mean Squared Difference (MSD), while applicable in many statistical applications, are not suitable in the field of RS, where not only are the data very sparse, but there is also a very small set of permitted values for the ratings.

Traditional metrics display a marked tendency to show high similarity between users based on the similarity of their ratings on a very small set of items. These metrics can assign maximum similarity to two users who have each rated hundreds of items but who have only rated three items in common.

Using the k-nearest neighbours (KNN) algorithm, it is common to find active users with a significant number of inadequate neighbours (neighbours who have little information in common with the active user). Our hypothesis is that it is possible to improve the quality of the recommendations of a CF RS if we use the Pareto dominance concept, which eliminates the less representative users from the k-neighbours selection process and keeps the most promising ones.

Fig. 1 shows the characteristics of the k-neighbours ($k = 150$) in the Movielens database and the positive impact of discarding neighbours with a small number of items in common. In this experiment, the following similarity metrics have been used: Pearson Correlation (COR), Cosine (COS), and Mean Squared Difference (MSD). Fig. 1A shows the percentage of items rated by the active user that was included in the calculation of the similarity metric (with each of the neighbours) using traditional CF. Clearly, traditional CF generally identifies the active user's neighbours using a very low percentage of his or her ratings. Fig. 1B displays a decreasing trend in the Mean Absolute Error (MAE) (that is, the improvement) when the neighbours are identified using a higher percentage of items in common with the active user. Fig. 1C displays an increasing trend in coverage when neighbours are identified using a higher percentage of items in common with the active user (the low coverage values arise from consideration of only a single neighbour). These experiments were repeated with the Netflix database, and very similar results were obtained.

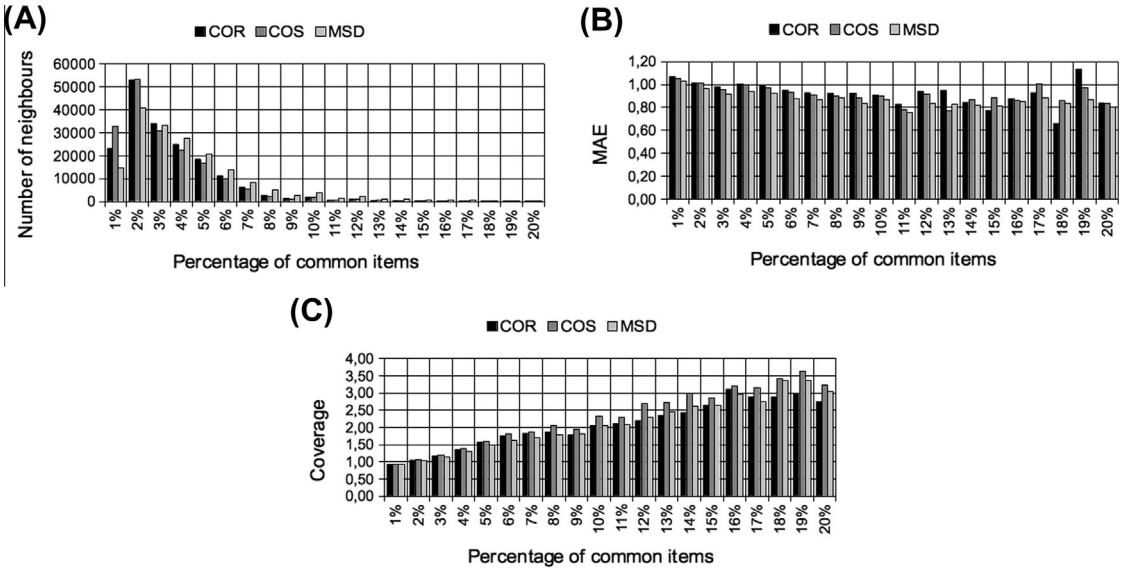

Fig. 1. Characteristics of the 150-neighbours in Movielens for different similarity metrics: Pearson Correlation (COR), Cosine (COS) and Mean Squared Difference (MSD). (A) Percentage of items rated by the active user that were used in the similarity metric calculation using traditional CF. (B) Trend in MAE when the percentage of common items between the active user and his or her neighbours is increased. (C) Trend in coverage when the percentage of common items between the active user and his or her neighbours is increased.

The proposed method attempts to solve this problem by using a novel approach, based on Pareto dominance, to exclude unpromising users in the *k*-neighbours selection phase.

2.3. The concept of dominance

Some authors have attempted to solve the problem stated in Section 2.2 by including a term in the similarity metric that discriminates against similarities calculated with too few items. In [9], the Jaccard Index is combined with the Mean Squared Difference to ensure that similar users have more items in common. In [10], the probability that the most important users (those with more knowledge about the items) are chosen as neighbours increases. In [16], the similarity metric is multiplied by a weighting factor corresponding to the percentage of common items that both the active user and the other user have rated.

The solution provided here is to use the Pareto Dominance concept used in multiobjective optimisation problems [30] to identify those users who correctly represent the user and who therefore must be considered to be candidate neighbours.

In a multiobjective optimisation problem, it is said that a solution x' is Pareto-optimal, efficient or non-dominated (under the minimisation hypothesis) if no other feasible solution exists (we will call the set of all feasible solutions X), which takes a lower value in some objective without causing a simultaneous increase in at least one other one. Formally, we are faced with a problem of optimisation in which we must find the vector $x = (x_1, \ldots, x_n)^T \in X$ that optimises the multiobjective function $f(x) = (f_1(x), \ldots, f_m(x))^T$.

We say that a solution $x' \in X$ is a non-dominated solution if there is no other solution $x \in X$ such that $f_i(x) \leqslant f_i(x')$ for all $i = 1, \ldots, m$ and there is at least one index i such that $f_i(x) < f_i(x')$.

The concept of dominance has been widely used in the design of algorithms to solve multiobjective optimisation problems. Among the most representative examples we find the algorithms NPGA [19], NSGA I [27] and II [13], SPEA2 [30], PAES [21], MOSA [29] and MOTS [15], which use dominance to make their solutions converge towards the Pareto-optimal.

In the field of RS, Pareto dominance has been used [28] to determine which items may be of interest to the user in a conversational recommender system. Conversational recommender systems are instances of content-based filtering and have been widely used in consumer-oriented e-commerce applications. In this paper we present a completely different approach that integrates Pareto dominance into the CF RS.

Fig. 2 shows a graphical representation of how the proposed method extends and complements traditional CF. Clearly, the proposed method does not modify the traditional method. Instead, it identifies a suitable set of initial users, ensuring that the traditional method obtains the most representative neighbours of the current user. The proposed method performs pre-processing which discards those users that do not provide information in addition to that provided by the finally selected set of users. An important advantage of the proposed method is that it is compatible with any other implementation or improvement of the traditional method, such as similarity measure improvement, and aggregation approach improvement.

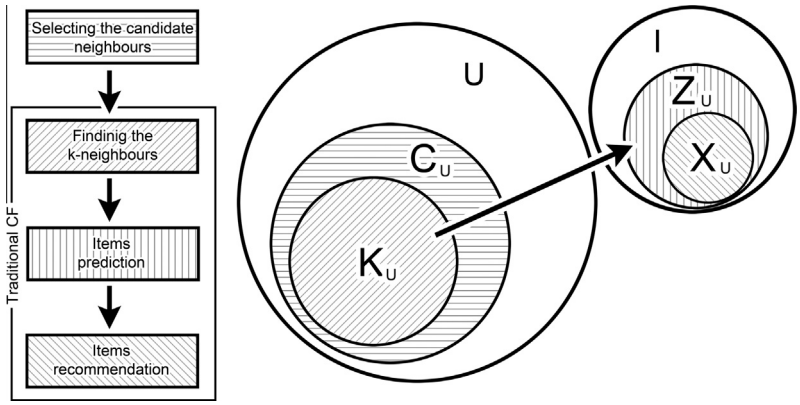

Fig. 2. Proposed method location and operation with respect to the traditional method. The mathematical notation used is the same as that used in Section 3.

3. Formalisation of the new method

3.1. Introduction

We define an RS based on CF having l users who choose ratings for m items from the interval $\{min, \ldots, max\}$, where the lack of a rating is represented by . The following sets are defined:

$$U = \{u \in \mathbb{N} | 1 \leqslant u \leqslant l\}, \text{ set of users} \tag{1}$$

$$I = \{i \in \mathbb{N} | 1 \leqslant i \leqslant m\}, \text{ set of items} \tag{2}$$

$$V = \{v \in \mathbb{N} | \min \leqslant v \leqslant \max\} \cup \{\bullet\}, \text{ set of possible ratings} \tag{3}$$

$$R_u = \{(i, v) | i \in I, v \in V\}, \text{ ratings of user } u \tag{4}$$

$$\text{We define the rating } v \text{ of the user } u \text{ on item } i \text{ as } r_{u,i} = v \tag{5}$$

$$\text{We define the ratings average of the user } u \text{ as } \bar{r}_u \tag{6}$$

We define the cardinality of a set C as its number of valid elements:

$$\#C = \#\{x \in C | x \neq \bullet\} \tag{7}$$

$$\#R_u = \#\{i \in I | r_{u,i} \neq \bullet\} \tag{8}$$

3.2. Selecting the candidate neighbours (non-dominated users)

We determine the set of users who are candidate neighbours of the active user or, more formally, the set of non-dominated users with respect to the active user.

$$\text{Let } I_u = \{i \in I | r_{u,i} \neq \bullet\} \text{ be the set of items rated by the user } u \tag{9}$$

Let $d(r_{x,i}, r_{y,i})$ be the absolute difference between the ratings given by user x and user y to the item i.

$$d(r_{x,i}, r_{y,i}) = \begin{cases} |r_{x,i} - r_{y,i}| & r_{y,i} \neq \bullet \\ \infty & r_{y,i} = \bullet \end{cases} \tag{10}$$

We say that user x dominates user y with respect to another user u (denoted as $x >_u y$), if the following expression (11) is satisfied.

$$x >_u y \iff \forall i \in I_u : d(r_{u,i}, r_{x,i}) \leqslant d(r_{u,i}, r_{y,i}) \wedge \exists j \in I_u | d(r_{u,j}, r_{x,j}) < d(r_{u,j}, r_{y,j}) \tag{11}$$

Conceptually, dominated users do not show greater similarity to the active user than the users that dominate them, but they do show lower similarity. In other words, dominated users do not provide any improvement compared to those that dominate them and can therefore be discarded.

We define C_u as the set of users who are candidate neighbours to the user u (non-dominated users). The following expression must be satisfied:

$$\text{Let } D_u \text{ be the set of users who are dominated by at least one user with respect to user } u \tag{12}$$

$$C_u \subset U, \quad u \notin C_u, \;\; C_u = U - (D_u \cup \{u\}), \;\; \forall y \in D_u, \;\; \exists x \in C_u | x >_u y \tag{13}$$

3.3. Finding the k-neighbours

To find the active user's k-neighbours we must complete the following steps:

1. Calculate the similarity of the active user to each of the users who are candidate neighbours (C_u).
2. Find the k users with the highest similarities to the active user.

The advantage of the proposed method arises from the fact that the process is applied to the subset $C_u \subset U$, to which the most suitable candidate neighbours belong, and not to the whole set of users of the RS, U, as in the traditional method.

To perform the first step we will use some of the traditional similarity measures [1,12]. For this paper, we have selected Pearson Correlation (15), Cosine (16) and Mean Squared Difference (17).

$$\text{Let } A_{x,y} = \{i \in I | r_{x,i} \neq \bullet \wedge r_{y,i} \neq \bullet\} \text{ be the set of items rated by both users } x \text{ and } y \tag{14}$$

Let x be the active user:

$$\text{correlation}(x,y) = \frac{\sum_{i \in A_{x,y}} (r_{x,i} - \bar{r}_x) \cdot (r_{y,i} - \bar{r}_y)}{\sqrt{\sum_{i \in A_{x,y}} (r_{x,i} - \bar{r}_x)^2 \cdot \sum_{i \in A_{x,y}} (r_{y,i} - \bar{r}_y)^2}} \Longleftrightarrow y \in C_x \tag{15}$$

$$\text{cosine}(x,y) = \frac{\sum_{i \in A_{x,y}} r_{x,i} \cdot r_{y,i}}{\sqrt{\sum_{i \in A_{x,y}} r_{x,i}^2} \cdot \sqrt{\sum_{i \in A_{x,y}} r_{y,i}^2}} \Longleftrightarrow y \in C_x \tag{16}$$

$$msd(x,y) = 1 - \frac{1}{\#A_{x,y}} \sum_{i \in A_{x,y}} \left(\frac{r_{x,i} - r_{y,i}}{\max - \min} \right)^2 \Longleftrightarrow y \in C_x \tag{17}$$

$$sim(x,y) = \bullet \Longleftrightarrow y \notin C_x \tag{18}$$

As we can see, the neighbours are chosen only from the set of non-dominated users.

For the second step, we define K_u as the set of k-neighbours of the active user. The following expressions must be true:

$$K_u \subseteq C_u \wedge \#K_u \leqslant k \wedge u \notin K_u \tag{19}$$

$$\forall x \in K_u, \quad \forall y \in (C_u - K_u) : sim(u,x) \geqslant sim(u,y) \tag{20}$$

Choosing the optimal value of k is not a straightforward task. The value of this parameter depends on several factors, including the size, sparsity and nature of the database, the singularity of the active user, and the type of items to be recommended. Brute force is required. For example, Schafer et al. [25] state "*...calculating a user's perfect neighbourhood is expensive – requiring comparison against all other users*." In this paper we have tested several values of the parameter k to approach an optimum for each experiment.

The candidate neighbours (C_u) should cover all of the active user's preferences (ratings). Therefore the number of neighbours required by each active user has a direct relationship to both the number of items rated by the user and the singularity of the user's ratings. In the first case, if the active user has rated more items, more candidates will be needed to cover all the user's preferences. In the second case, if the ratings made are very unusual, it will be very difficult to find users that represent them (the required number of candidate neighbours will be higher), whereas if the ratings are very common, fewer users will be needed to represent them (the number of candidate neighbours will be lower).

Fig. 3 shows the relationship between the number of candidate neighbours determined for active users and the number of items they have rated in the Movielens database. The quotient between the number of candidates and the number of ratings defines an active user's singularity. If the quotient is small, few users will be needed to cover all of the active user's preferences, and, therefore, the active user is not very singular. On the other hand, if the quotient is large, it means many users were needed to cover all of the active user's preferences; therefore, the active user is very singular.

The Gaussian distribution displayed in Fig. 3 indicates the balance provided by the proposed singularity measure, which is derived from the "set of candidates (C_u)" concept given in this paper.

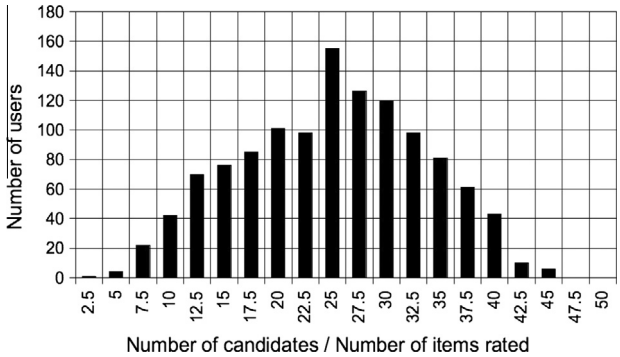

Fig. 3. Relationship between the number of candidate neighbours and the number of ratings made by active users in the Movielens database.

3.4. Item recommendation

The process by which recommendations are made to the active user can be divided into two steps:

1. Determine the rating predictions for the active user based on ratings made by the set of k-neighbours.
2. Find the N items with the highest predictions and recommend these items to the active user.

To complete the first step (prediction), we must define the way in which the ratings of the k-neighbours are combined. Traditionally, we have the following aggregation approaches [1,9,12]: mean (22), weighted mean (23) and deviation from the mean (24).

$$\text{Let } G_{u,i} = \{n \in K_u | r_{n,i} \neq \bullet\} \text{ be the set of neighbours who have rated item } i \tag{21}$$

Let $p_{u,i}$ be the prediction of item i to user u:

$$p_{u,i} = \frac{1}{\#G_{u,i}} \sum_{n \in G_{u,i}} r_{n,i} \Longleftrightarrow G_{u,i} \neq \emptyset \tag{22}$$

$$p_{u,i} = \frac{\sum_{n \in G_{u,i}} sim(u,n) \cdot r_{n,i}}{\sum_{n \in G_{u,i}} sim(u,n)} \Longleftrightarrow G_{u,i} \neq \emptyset \tag{23}$$

$$p_{u,i} = \bar{r}_u + \frac{\sum_{n \in G_{u,i}} sim(u,n) \cdot (r_{n,i} - \bar{r}_n)}{\sum_{n \in G_{u,i}} sim(u,n)} \Longleftrightarrow G_{u,i} \neq \emptyset \tag{24}$$

$$p_{u,i} = \bullet \Longleftrightarrow G_{u,i} = \emptyset \tag{25}$$

To develop the second step (recommendation), we define X_u as the set of items likely to be recommended to user u, and Z_u is defined as the set of (at most) N items to be recommended. The following expressions must be satisfied:

$$X_u \subset I \wedge \forall i \in X_u, \quad r_{u,i} = \bullet, \;\; p_{u,i} \neq \bullet \tag{26}$$

$$Z_u \subseteq X_u, \quad \#Z_u \leqslant N, \;\; \forall x \in Z_u, \;\; \forall y \in (X_u - Z_u) : p_{u,x} \geqslant p_{u,y} \tag{27}$$

3.5. Sample computation

To better understand how the proposed method performs, we have developed a sample computation. Table 1 shows the parameters used in this sample computation. These parameters have been selected with the purpose of simulating a small, easily understandable CF RS. We define a traditional rating range (one to five stars) and a reduced number of users, items, neighbours and recommendations.

The users' ratings of the items are displayed in Table 2.

Our objective is to determine the recommendations to make for User 1 (U_1). First we find the set of candidate neighbours, i.e., those users who are non-dominated with respect to User 1. These relationships are displayed in Table 3, in which the first two columns ($d^i_{x,y}$ *and* U_1 *rated items*) contain the differences between User 1's ratings and those of the rest of user. The third

Table 1
Parameters used in the sample computation.

Parameter	Value
Number of users (l)	6
Number of items (m)	10
Number of neighbours (k)	2
Number of recommendations (N)	2
Maximum vote (max)	5
Minimum vote (min)	1

Table 2
Users' ratings of items in the sample computation.

$r_{u,i}$	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}
U_1	•	3	5	•	2	4	•	1	5	•
U_2	4	3	4	•	•	4	1	2	•	•
U_3	5	1	5	4	2	•	3	•	4	•
U_4	•	4	3	2	•	1	•	5	•	•
U_5	•	•	3	4	5	•	2	•	3	2
U_6	•	5	2	•	•	5	3	1	5	4

Table 3
Candidate neighbours of U_1 in the sample computation.

$d^i_{x,y}$	U_1 rated items						Dominated by	Candidates (C_{U1})
	I_2	I_3	I_5	I_6	I_8	I_9		
U_2	0	1	∞	0	1	∞	Ø	$U - (\{U_4, U_5\} \cup \{U_1\}) = \{U_2, U_3, U_6\}$
U_3	2	0	0	∞	∞	1	Ø	
U_4	1	2	∞	3	4	∞	$\{U_2\}$	
U_5	∞	2	3	∞	∞	2	$\{U_3\}$	
U_6	2	3	∞	1	0	0	Ø	

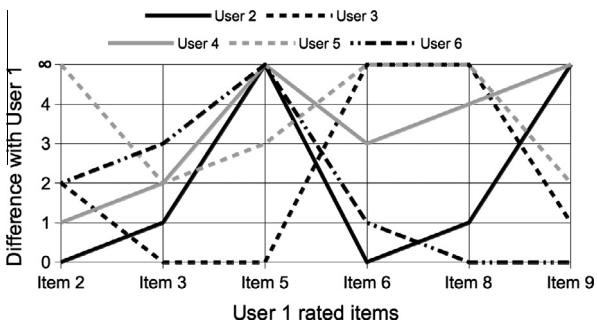

Fig. 4. Graphical representation of the difference in ratings between User 1 and the other users of the system, for calculating the dominating and dominated users with respect to User 1.

Table 4
Similarities between U_1 and the set of candidate neighbours (C_{U1}) and set of two nearest neighbours obtained.

	Mean squared differences (16)					Neighbours (K_{U1})
	U_2	U_3	U_4	U_5	U_6	
U_1	0.937	0.859	•	•	0.55	$\{U_2, U_3\}$

Table 5
Recommendations made to U_1 in the sample computation.

	I_1	I_4	I_7	I_{10}	X_{U1}	Z_{U1}
$p_{U1,i}$	4.5	4.0	2.0	•	$\{I_1, I_4, I_7\}$	$\{I_1, I_4\}$

column (*Dominated by*) contains the users who dominate other users with respect to User 1. The last column contains User 1's candidate neighbours (C_{U1}).

Fig. 4 displays this process graphically. The horizontal axis shows the items rated by User 1, and the vertical axis shows the absolute difference between User 1's ratings and those of the rest of the users. Gray lines represent dominated users and black lines represent non-dominated users. For example, we can see that User 2 dominates User 4 because the line representing User 2 is always below or at the same height as the line representing User 4. User 3 dominates User 5 for the same reason.

Table 4 shows the set of k-neighbours selected from the set of candidates. We use MSD (17) to measure the similarity between users. Those users who do not belong to the set of candidates of U_1 have no similarity to them (18).

Table 5 indicates the predictions made for items not rated by U_1, using the arithmetic mean (22) as the aggregation approach. X_u refers to the set of items likely to be recommended. Z_u refers to the recommendations actually made.

4. Experimental design

4.1. Quality measures

To validate the behaviour of the proposed method, we use the following prediction and recommendation quality measures: mean absolute error, coverage, precision and recall [5,16–18].

We use the Mean Absolute Error (MAE) to determine the mean error that occurs in the aggregate of all the predictions made. We define a user's MAE as follows:

$$\text{Let } B_u = \{i \in I | r_{u,i} \neq \bullet \wedge p_{u,i} \neq \bullet\} \text{ be the set of items rated by user } u \text{ where a prediction can be obtained} \tag{28}$$

$$mae_u = \frac{1}{\#B_u} \sum_{i \in B_u} |r_{u,i} - p_{u,i}| \tag{29}$$

We define the system's MAE as follows:

$$mae = \frac{1}{\#U} \sum_{u \in U} mae_u \tag{30}$$

We use the coverage to measure the capacity of a user's k-neighbours to recommend new items. We define the user's coverage as follows:

$$\text{coverage}_u = 100 \times \frac{\#\{i \in I | r_{u,i} = \bullet \wedge p_{u,i} \neq \bullet\}}{\#\{i \in I | r_{u,i} = \bullet\}} \tag{31}$$

and the system's coverage as follows:

$$\text{coverage} = \frac{1}{\#U} \sum_{u \in U} \text{coverage}_u \tag{32}$$

We use the precision and the recall to measure the quality of the recommendations made. The precision indicates the percentage of relevant recommended items with respect to the total number of items recommended. The recall indicates the percentage of relevant recommended items with respect to the total number of relevant items. An item is considered relevant or not relevant according to the parameter θ. To calculate the precision and the recall, we re-define (26) and (27) as follow:

$$X'_u \subset I \wedge \forall i \in X'_u, \quad r_{u,i} \neq \bullet, \ \ p_{u,i} \neq \bullet \tag{33}$$

$$Z'_u \subseteq X'_u, \quad \#Z'_u \leqslant N, \ \ \forall x \in Z'_u, \ \ \forall y \in (X'_u - Z'_u) : p_{u,x} \geqslant p_{u,y} \tag{34}$$

We define a user's precision as follows:

$$\text{precision}_u = \frac{\#\{i \in Z'_u | r_{u,i} \geqslant \theta\}}{\#Z'_u} \tag{35}$$

and the system's precision as follows:

$$\text{precision} = \frac{1}{\#U} \sum_{u \in U} \text{precision}_u \tag{36}$$

We define a user's recall as follows:

$$\text{recall}_u = \frac{\#\{i \in Z'_u | r_{u,i} \geqslant \theta\}}{\#\{i \in I | r_{u,i} \geqslant \theta\}} \tag{37}$$

Table 6
Main parameters of the databases used in the experiments.

	Movielens	Netflix
Number of users	4382	480189
Number of items	3952	17,770
Number of rates	1000209	100480507
Min and max values	1–5	1–5

Table 7
Experiment parameters.

	Prediction		Recommendation			Cross-validation		Figures
	k	Step k	N	k	θ	Users (%)	Items (%)	
Movielens	$\{50,\ldots,800\}$	50	$\{1,\ldots,20\}$	100	5	20	20	Fig. 6
Netflix	$\{50,\ldots,800\}$	50	$\{1,\ldots,20\}$	200	5	5	20	Fig. 7

and the system's recall as follows:

$$\text{recall} = \frac{1}{\#U}\sum_{u\in U}\text{recall}_u \tag{38}$$

4.2. Experiments performed

The standard RS baseline is k-nearest neighbours CF, using Pearson Correlation as the similarity measure and deviation from the mean as the aggregation approach [5]. As the proposed method is a pre-processing step applied prior to classical CF, in our experiments we determine whether this step improves the CF RS results. If our hypothesis is correct, a significant improvement in all quality measures should be observed.

In our experiments we use Pearson Correlation (15), Cosine (16) and MSD (17) as similarity measures. As an aggregation approach we use deviation from the mean (24). One test is performed using the traditional method and one using the proposed method. We then determine whether any improvement is observed in the above-mentioned quality measures: MAE (30), coverage (32), precision (36) and recall (38).

The experiments are performed on the Movielens and Netflix databases, the main parameters of which can be seen in Table 6. All the experiments are performed using cross-validation. In the case of Movielens we use 20% of test users to perform the experiments, whereas in Netflix we use 5% of test users. In both cases, the percentage of test items is 20%.

Table 7 shows the main parameters used in the experiments, the databases tested, and the figures where the results are presented.

5. Results

In this section, we present the results obtained from the experiments using the Movielens and Netflix databases (Table 6) using the parameters listed in Table 7.

Fig. 5 shows the percentage of items rated by the active user that were used in the calculation of the similarity metric (with each of the neighbours) using the proposed method (Pre-Filtered metrics). The following similarity metrics are shown: Pearson correlation (COR), Cosine (COS) and Mean Squared Difference (MSD). Compared with traditional CF (Fig. 1A), the proposed method generally identifies the neighbours based upon a much higher number of items in common, which guarantees better knowledge of the real similarity between each pair of users.

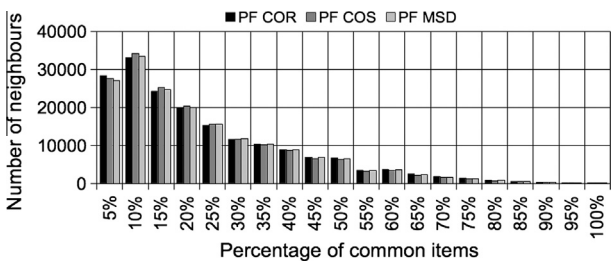

Fig. 5. Percentage of the items rated by the active user that were used in the similarity metric calculation using the proposed CF.

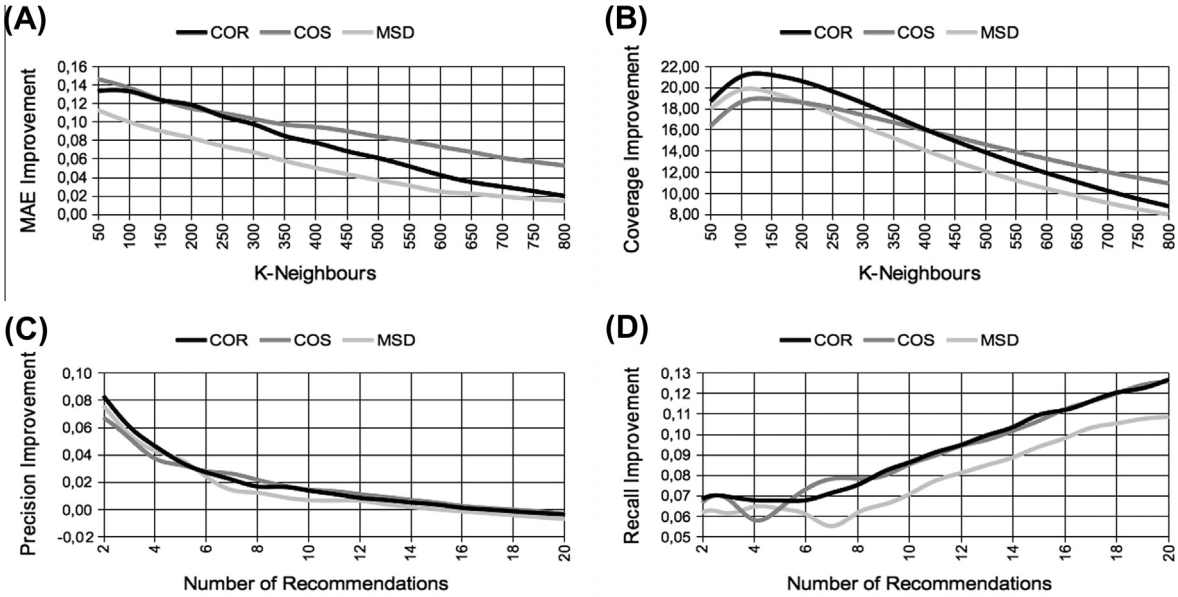

Fig. 6. Improvements produced in quality measures results using the proposed method instead of the traditional method. Database: Movielens. (A) MAE. (B) Coverage. (C) Precision. (D) Recall.

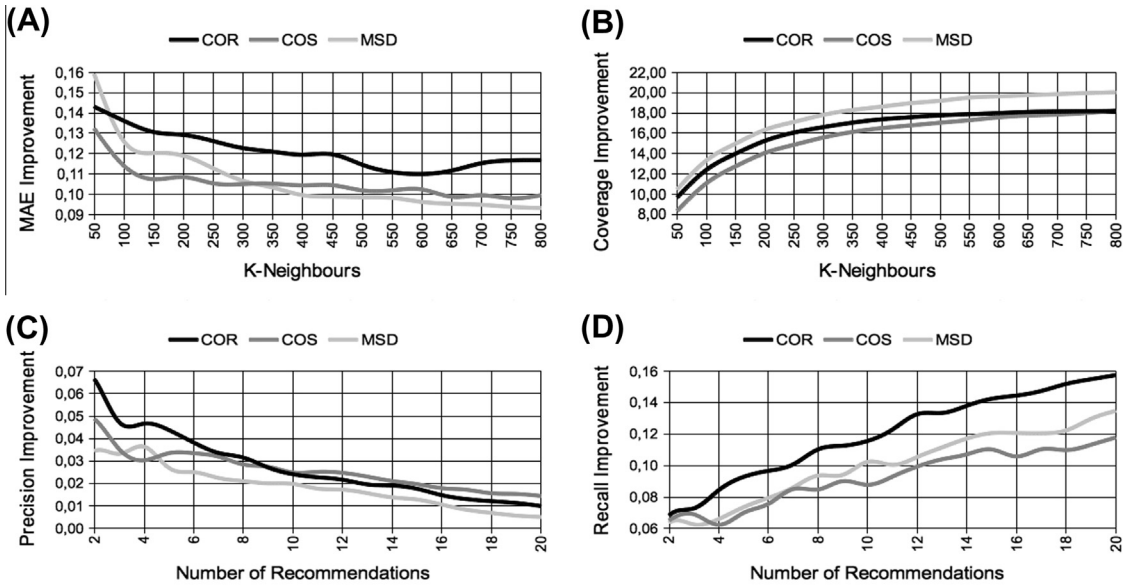

Fig. 7. Improvements produced in quality measures results using the proposed method instead of the traditional method. Database: Netflix. (A) MAE. (B) Coverage. (C) Precision. (D) Recall.

Fig. 6 shows the improvement produced in the quality measures for the Movielens database when the proposed method is used rather than the traditional method. Fig. 6A shows the improvement produced in the quality of the predictions (MAE). The most notable improvement is obtained with small values of k because as k increases, the neighbourhoods of the two approaches tend to be more and more similar and the quality of the predictions tends to be the same. Fig. 6B shows an increase in the coverage using the proposed method. For this measure, the most notable improvements are also produced for small values of k. This result is important because we achieve good prediction quality in the interval of k values (low values) where the operation of RS are more efficient.

Figs. 6C and D show the improvement produced in precision and recall, respectively. For these experiments, we used a value of $k = 100$ and a relevance threshold of $\theta = 5$. The proposed method always produces an improvement. This

improvement decreases with the number of recommendations in the precision and increases with the number of recommendations in the recall.

Fig. 7 displays the improvement obtained using the proposed method with the Netflix database. Fig. 7A shows the improvement in MAE, which is more consistent with the Netflix database than with the Movielens database. This improvement arises because the number of Netflix users is much greater than the number of Movielens users, which allows the proposed method to discard a higher number of dominated users while simultaneously including a higher number of dominating users.

Fig. 7B shows that coverage increases as k increases. Using the Netflix database, it becomes necessary to achieve a balance between accuracy and coverage. In contrast to the results obtained using the Movielens database, with Netflix the coverage improvement increases as the number of neighbours increases. This is because the number of items in Netflix is much larger; therefore, it is easier to recommend items not rated by the active user [5–7].

Figs. 7C and D show the improvement produced in precision and recall, respectively. For these experiments, we used a value of $k = 200$ and a relevance threshold of $\theta = 5$. The trends are similar to those observed using the Movielens database; therefore, we find that there is a general improvement in recommendations obtained using the proposed method compared with those obtained using the traditional method.

6. Conclusions and future work

The traditional collaborative filtering method selects insufficiently representative users as neighbours of the active user. This means that recommendations made *a posteriori* are not sufficiently precise. The method proposed in this paper involves a pre-filtering process that eliminates the least representative users from the k-neighbour selection process and retains the most promising ones.

The improvements obtained are always positive with respect to both the prediction quality measures and the recommendation quality measures. This demonstrates that certain users should not be included among the active user's neighbours and that the traditional similarity measures are not capable of detecting them. The favourable results obtained here can be considered generally applicable due to the broad margin of improvement observed and their testing on the two most representative databases of collaborative filtering recommender systems.

The proposed method is appropriate for use in all collaborative filtering recommender systems, as it requires only the introduction of one additional (prior) phase to the traditional method. The proposed method is applicable to all memory-based collaborative filtering recommender systems, is easy to implement and is effective in producing improved results.

For future work, we propose making use of the concept of a user's singularity, which is obtained as the proportion between the number of candidates (C_u) and the number of items rated by the active user u, as shown in Fig. 3. This singularity concept can be used to improve the accuracy of the RS or as a parameter to define a collaborative filtering trust measure.

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